

SIJIL PELAJARAN MALAYSIA • MATEMATIK (1449)
SKEMA PEMARKAHAN LENGKAP / COMPLETE MARKING SCHEME

Peperiksaan Ramalan / Percubaan SPM 2026

KERTAS 1 (1449/1) — SKEMA JAWAPAN

No	Jaw	No	Jaw	No	Jaw	No	Jaw
1	A	11	A	21	B	31	B
2	C	12	B	22	A	32	B
3	A	13	C	23	C	33	B
4	A	14	B	24	A	34	B
5	A	15	C	25	A	35	C
6	B	16	A	26	A	36	D
7	C	17	C	27	A	37	A
8	A	18	B	28	A	38	B
9	A	19	A	29	A	39	B
10	B	20	B	30	B	40	A

Soalan 1 (Bahagian A)

[3 markah / 3 marks]

(a) Tukar ke asas 10: $110101_2 = 53_{10}$, $324_5 = (3 \times 25) + (2 \times 5) + 4 = 89_{10}$ K1

$p_8 = 89 - 53 = 36_{10} \implies 36_{10} = (4 \times 8) + 4 = 44_8 \implies p = 44$ N1

(b) Nilai digit 4 = $4 \times 5^2 = 100$ N1

Marker's Tip: State $p = 44$ (the digits in base 8), not 44_8 as the variable value.

Soalan 2 (Bahagian A)

[3 markah / 3 marks]

(a) Palsu / False (kerana $(-4)^2 = +16$, bukan -16) N1

(b) Premis 2: Sebuah heptagon sekata ialah poligon sekata dengan 7 sisi / A regular heptagon is a regular polygon with 7 sides. N1

(c) $2n^2 + 1$, $n = 1, 2, 3, \dots$ N1

Marker's Tip: For inductive conclusion, always state the domain $n = 1, 2, 3, \dots$ to receive the full mark.

Soalan 3 (Bahagian A)

[3 markah / 3 marks]

(a) Kawasan di dalam A tetapi di luar B : $3 + 2 = 5$ N1

(b) $(A \cup B) \cap C' = 3 + 5$ K1

$= 8$ N1

Marker's Tip: Region $(A \cup B) \cap C'$ refers to elements inside A or B that are strictly outside circle C .

Soalan 4 (Bahagian A)

[4 markah / 4 marks]

Kekangan I: $x + y \leq 80$ K1

Kekangan II: $y \geq \frac{1}{2}x$ atau $2y \geq x$ K1

Kekangan III: $20x + 30y \geq 1200$ atau $2x + 3y \geq 120$ K1

Ketiga-tiga ketaksamaan ditulis dengan tepat tanpa ralat simbol. N1

Marker's Tip: 'Tidak melebihi' translates to $\leq$, while 'sekurang-kurangnya' translates to $\geq$.

Soalan 5 (Bahagian A)

[4 markah / 4 marks]

(a) Dari $2y - 4x = 7 \implies y = 2x + 3.5 \implies m_{PQ} = 2 \implies m_{RS} = 2$ N1

(b) $y - y_1 = m(x - x_1) \implies y - 2 = 2(x - 3) \implies y - 2 = 2x - 6$ K1

$y = 2x - 4$ N1

(c) Pada pintasan- x , $y = 0 \implies 0 = 2x - 4 \implies x = 2$ N1

Marker's Tip: For x -intercept, substitute $y = 0$ into the linear equation.

Soalan 6 (Bahagian A)**[4 markah / 4 marks]**

(a) Tempoh laju seragam = 10 s (atau 9 s dari $t = 16$ ke $t = 25$)

N1

(b) Kadar perubahan laju = $\frac{12-12}{25-16} = 0 \text{ m s}^{-2}$

N1

(c) Jumlah jarak = $(10 \times 30) + \frac{1}{2}(30 + 12)(6) + (9 \times 12) = 300 + 126 + 108$
= 534 m

K1

N1

Marker's Tip: Split the area under the graph into Rectangle 1, Trapezium, and Rectangle 2.

Soalan 7 (Bahagian A)**[4 markah / 4 marks]**

(a) Kos selepas deduktibel = $24400 - 400 = \text{RM}24,000$; Ditanggung insurans = 0.85×24000

K1

= RM20,400

N1

(b) Bahagian ko-insurans Encik Amirul = $0.15 \times 24000 = \text{RM}3,600$; Jumlah bayaran = $400 + 3600$

K1

= RM4,000

N1

Marker's Tip: The deductible is always deducted first before applying the 85/15 ratio.

Soalan 8 (Bahagian A)**[5 markah / 5 marks]**

(a) $W = \frac{k\sqrt{x}}{y^2} \implies 6 = \frac{k\sqrt{16}}{2^2} \implies 6 = \frac{4k}{4} \implies k = 6$

K1

$W = \frac{6\sqrt{x}}{y^2}$

N1

(b)(i) $W = \frac{6\sqrt{36}}{3^2} = \frac{6(6)}{9} = 4$

N1

(b)(ii) $2 = \frac{6\sqrt{64}}{y^2} \implies 2y^2 = 6(8) = 48 \implies y^2 = 24$

K1

$y = \sqrt{24} = 4.899 \text{ atau } 2\sqrt{6}$

N1

Marker's Tip: Question specifies 'positive value of y ', so reject negative root $-\sqrt{24}$.**Soalan 9 (Bahagian A)****[5 markah / 5 marks]**

(a) $\sum x = N\bar{x} = 5 \times 8 = 40$

P1

$\sigma^2 = \frac{\sum x^2}{N} - \bar{x}^2 \implies 6.4 = \frac{\sum x^2}{5} - 64 \implies \sum x^2 = 5(70.4) = 352$

N1

(b)(i) $\bar{x}_{\text{baru}} = \frac{40+14}{6} = \frac{54}{6} = 9$

N1

(b)(ii) $\sum x_{\text{baru}}^2 = 352 + 14^2 = 352 + 196 = 548$; $\sigma = \sqrt{\frac{548}{6} - 9^2} = \sqrt{91.333 - 81} = \sqrt{10.333}$

K1

= 3.21

N1

Marker's Tip: The new number of observations N increases from 5 to 6.

Soalan 10 (Bahagian A)

[5 markah / 5 marks]

(a) $S = \{(M, 2), (M, 3), (M, 5), (A, 2), (A, 3), (A, 5), (T, 2), (T, 3), (T, 5), (H, 2), (H, 3), (H, 5)\}$, $n(S) = 12$ N1

(b)(i) Konsonan $\{M, T, H\}$, Perdana $\{2, 3, 5\} \implies 3 \times 3 = 9$ kesudahan $\implies P = \frac{9}{12} = \frac{3}{4}$ N1

(b)(ii) Vokal atau Ganjil: $\{(A, 2), (A, 3), (A, 5), (M, 3), (M, 5), (T, 3), (T, 5), (H, 3), (H, 5)\} \implies 9$ kesudahan K1

$P = \frac{9}{12} = \frac{3}{4}$ N1

Marker's Tip: List all outcomes in set notation before calculating probability.

Soalan 11 (Bahagian B)

[9 markah / 9 marks]

(a)(i) Jarak titik $K(4, 1)$ ke garis $x = 2$ ialah 2 unit ke kanan $\implies$ Imej $V(K) = (0, 1)$ N1

(a)(ii) Putaran $(0, 1)$ 90° ikut arah jam pada pusat $(1, 3)$: Vektor $\begin{pmatrix} -1 \\ -2 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \end{pmatrix} = (-1, 4)$ K1

$UV(K) = (-1, 4)$ N1

(b)(i) Faktor skala $k = \frac{EF}{AB} = \frac{9}{3} = 3$ N1

(b)(ii) Luas imej $EFGH = 3^2 \times 14 = 9 \times 14 = 126 \text{ cm}^2$ K1

Luas kawasan berllorek $= 126 - 14 = 112 \text{ cm}^2$ N1

(c) Bentuk dan saiz yang serupa secara persis (semua panjang sisi sepadan dan sudut sepadan adalah sama). P2

Marker's Tip: UV means perform transformation V first, followed by transformation U .

Soalan 12 (Bahagian B)

[9 markah / 9 marks]

(a) Dongakan Depan X : Bentuk poligon $ABLKGF$ dengan ukuran tepat $AB = 6 \text{ cm}$, $AF = 5 \text{ cm}$, $FG = 2 \text{ cm}$, $LK = 2 \text{ cm}$, $CK = 3 \text{ cm}$ K2

Semua garis lurus kemas dan bersambung tepat $\pm 0.1 \text{ cm}$ N1

(b) Dongakan Sisi Y : Segi empat tepat $8 \text{ cm} \times 5 \text{ cm}$ dengan garis sempadan aras 3 cm dilukis tepat. K2

Sudut tepat 90° dan ukuran tepat. N1

(c) Pelan: Segi empat tepat $6 \text{ cm} \times 8 \text{ cm}$ dengan satah atas FG , serong, dan LK dibahagikan mengikut ukuran sebenar. K2

Ketepatan garisan dan skala penuh sempurna. N1

Marker's Tip: Use solid thick lines for visible edges and dashed lines for hidden interior edges.

Soalan 13 (Bahagian B)

[9 markah / 9 marks]

(a) $|A| = (5)(1) - (2)(3) = 5 - 6 = -1 \implies A^{-1} = \frac{1}{-1} \begin{pmatrix} 1 & -2 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix}$ N1

(b)(i) $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 19 \\ 11 \end{pmatrix}$ P2

(b)(ii) $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 3 & -5 \end{pmatrix} \begin{pmatrix} 19 \\ 11 \end{pmatrix} = \begin{pmatrix} -1(19) + 2(11) \\ 3(19) - 5(11) \end{pmatrix}$ K1

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \implies$ Harga sebatang pen = RM3.00, Harga sebuah buku nota = RM2.00 N2

(b)(iii) Jumlah kos = $10(3) + 4(2) = 30 + 8 = \text{RM}38.00$ N1

Marker's Tip: You MUST show the matrix multiplication explicitly to gain the method mark K1.

Soalan 14 (Bahagian B)

[9 markah / 9 marks]

(a) Sempadan atas: 30.5, 40.5, 50.5, 60.5, 70.5, 80.5; Kekerapan longgokan: 4, 12, 26, 44, 54, 60 P1

(b) Skala seragam pada kedua-dua paksi dan 7 titik diplot dengan tepat bermula pada (20.5, 0) K2

Lengkung ogif licin dan berterusan melalui semua titik. N1

(c)(i) Median (pada $N/2 = 30$) $\approx 52.5 \pm 1.0$ N1

(c)(ii) Q_1 (pada 15) ≈ 42.5 , Q_3 (pada 45) $\approx 61.0 \implies \text{IQR} = 61.0 - 42.5 = 18.5$ N2

(d) Pada skor 65, kekerapan longgokan = 49 $\implies$ Bilangan layak = $60 - 49 = 11 \implies \frac{11}{60} \times 100\% = 18.33\%$ N1

Marker's Tip: Ogive must always start on the horizontal axis at the lower boundary of the first class (20.5, 0).

Soalan 15 (Bahagian B)

[9 markah / 9 marks]

(a)(i) Pokok minimum mengandungi $V - 1 = 5$ tepi terpilih: $(R, S) = 4$, $(Q, U) = 5$, $(S, Q) = 6$, $(P, Q) = 7$, $(P, T) = 9$ K1

(a)(ii) Jumlah pemberat minimum = $4 + 5 + 6 + 7 + 9 = 31 \text{ km}$ N1

(b)(i) Pendapatan bercukai = $82000 - 2000 - 23000 = \text{RM}57,000$ N2

(b)(ii) Cukai = $1800 + 0.13(57000 - 50000) = 1800 + 0.13(7000) = 1800 + 910 = \text{RM}2,710$ K2

Jumlah cukai pendapatan perlu dibayar = RM2,710 N1

(b)(iii) Jumlah PCB tahunan = $12 \times 250 = \text{RM}3,000 > \text{RM}2,710 \implies$ **Menerima bayaran balik cukai sebanyak RM290** ($3000 - 2710$) N1

Marker's Tip: PCB refund occurs when annual PCB deduction is greater than the actual calculated tax.

Soalan 16 (Bahagian C)

[15 markah / 15 marks]

(a)(i) $P(15) = -2(15)^2 + 80(15) - 300 = -2(225) + 1200 - 300 = -450 + 1200 - 300 =$
RM450

N1

(a)(ii) $x = -\frac{b}{2a} = -\frac{80}{2(-2)} = 20 \implies 20 \times 10 = 200$ pasu cili

K1

$P(20) = -2(400) + 80(20) - 300 = -800 + 1600 - 300 =$ RM500

N1

(b) Tepi pokok minimum: $(S, M_1) = 6$ km, $(M_1, M_2) = 5$ km, $(M_1, M_3) = 8$ km, $(M_3, M_4) = 4$ km

K3

Jumlah jarak minimum $= 6 + 5 + 8 + 4 = 23$ km

N1

(c)(i) Aliran tunai bulanan $= 2400 - 1600 = +$ RM800

N1

(c)(ii) Simpanan bulanan diperlukan $= \frac{3600}{12} =$ RM300. Kerana $\text{RM}300 \leq \text{RM}800$, matlamat ini **Boleh Dicapai (Achievable/SMART)**.

K2

(d) Kaedah *A* lebih konsisten kerana sisihan piawainya (2.1 g) lebih kecil daripada Kaedah *B* (5.8 g), menunjukkan serakan data yang lebih rendah.

N2

Marker's Tip: Consistency in statistics is always determined by a smaller standard deviation.

Soalan 17 (Bahagian C)

[15 markah / 15 marks]

(a)(i) Luas panel solar $= 10 \times 5 = 50$ m²

N1

(a)(ii) Kenaikan tegak $= 5 \sin 30^\circ = 5(0.5) = 2.5$ m

K1

Tinggi maksimum dari tanah $= 4 \text{ m} + 2.5 \text{ m} = 6.5$ m

N2

(b)(i) Luas poster $= (2.5)^2 \times 50 = 6.25 \times 50 = 312.5$ cm²

N1

(b)(ii) Pembesaran pada pusat asalan $(0, 0)$ dengan faktor skala 2.5.

P2

(c) Jumlah insurans harus dibeli $= 0.80 \times 600000 = \text{RM}480,000$; Pampasan $= \frac{400000}{480000} \times 48000 -$
 $3000 = 40000 - 3000$

K2

$= \text{RM}37,000$

N1

(d)(i) Rajah pokok lengkap menunjukkan 4 cabang keberhasilan: $(C_1, C_2), (C_1, C'_2), (C'_1, C_2), (C'_1, C'_2)$

P1

(d)(ii) $P(\text{sekurang-kurangnya satu berfungsi}) = 1 - P(C'_1 \cap C'_2) = 1 - (0.05 \times 0.10) = 1 -$
 $0.005 = 0.995$

N2

Marker's Tip: Co-insurance compensation formula: $\frac{\text{Amount insured}}{\text{Required insurance}} \times \text{Loss} - \text{Deductible}$.